3. (i) Find

$$\int \frac{12}{(2x-1)^2}\,\mathrm{d}x$$

giving your answer in simplest form.

(2)

(ii) (a) Write $\dfrac{4x+3}{x+2}$ in the form

$$A + \frac{B}{x+2}$$ where A and B are constants to be found

(b) Hence find, using algebraic integration, the exact value of

$$\int_{-8}^{-5} \frac{4x+3}{x+2}\,\mathrm{d}x$$

giving your answer in simplest form.

(6)